MILITARY SPECIFICATION

TEST SET, ATOMIC WEAPON: T4090

1.0 SCOPE

1. This specification covers special requirements, quality assurance provisions and the preparation for delivery criterion for one type of test set used on XM-454 Projectile.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.

MILITARY

MIL-B-2427 - Box, Ammunition Packing, Wood, Nailed.
MIL-C-50228 (MU) Case, Transit for Test Set: T4090.
MIL-G-50229 (MU) Generator, Hand Crank.
MIL-P-116 Preservation, Methods of.
MIL-R-11/8 - Resistor Fixed Composition (Insulated), Style RC07.
MIL-S-60902 (MU) - Soldering Electrical Connections for Nuclear Weapons Systems, Procedure For

STANDARDS

MILITARY

AN 340 - Nut
AN 507 - Screw
AN 515 - Screw
AN 535 - Screw, drive

FSC 1190

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-170 - Moisture Resistance Test Cycle for Ground Signal Equipment.
MIL-STD-1168 (MU) Lot Numbering of Ammunition
MS 28775 - Packing "O" Ring
MS 35043 - Resister
MS 35233 - Screw
MS 51025 - Set Screw
MS 51041 - Set Screw
MS 122116 - Insert Heli-Coil
NAS 1201C6A - Coupling
NAS 1201CB10A - Coupling and Chain Assembly

DRAWINGS

ORDNANCE CORPS

8824530 - Silicon Zener Regulator, 1N1799
8824531 - Silicon Diode, 1N647
8824548 - Lamp, Neon Glow
8824549 - Terminal
8824556 - Connector, Receptacle, Electrical
8834253 - Connector, Electrical
8834268 - Epoxy Resin Top Coat
8834269 - Epoxy Resin, Putty Blend
8834301 - Generator, Hand Crank
8834337 - Plate, Cover Generator
8834340 - Module Assembly, Indicator Light
8834341 - Plate, Retaining, Module
8834345 - Crank Assy
8834346 - Test Set, Atomic Weapon T4090
8834347 - Plate Connector
8834349 - Housing, Test Set, Atomic Weapon T4090
8834386 - Yoke, Crank
8834388 - Arm Crank
8834394 - Plunger Ball
8834406 - Connector, Electrical, Modified
8834408 - Module Assy, Component, Electrical
8834409 - Wiring Diagram, Test Set T4090
8834410 - Schematic, Test Set T4090
8834440 - Electrofilm No 4856
8834465 - Identification Plate, Test Set Atomic Weapon T4090
8834470 - Barrier Light
8834471 - Face Plate, Top
8834472 - Face Plate, Bottom

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8834489 - Body, Module, Indicator Light
8834523 - Electrofilm No. 99A
8834564 - Connector Assembly
8834979 - Test Set, Atomic Weapon T4090 W/Case
8834981 - Case, Transit
8844885 - Box, Pkg, Ammo, For Test Set: T4090
ET -8834346 - Equipment Tabulation

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. - The contractor shall furnish objective evidence (see 6.2) that the material used complies with the specifications listed on the applicable drawing.

3.2 Assembly.- The test set shall comply with all requirements specified on Drawing (dwgs) 8834979, and 8834346 with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawing and herein.

3.2.1 Hand crank generator (see dwg. 8834301). - The hand crank generator shall comply with all the requirements of MIL-G-50229(MU).

3.2.2 Case Transit (see dwg 8834981) - The transit case shall comply with MIL-C-50228(MU).

3.3 First article testing.- Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization . - The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Circuits. - The circuits in the test set shall comply with the requirements of Table I. Tests will be conducted as specified in 4.4.1.

TABLE I

Circuit Requirements

1. The resistance (K ohms) between pins 3 and 9 shall be 22 ± 2.2.
2. The resistance (K ohms) between pins 9 and 11 shall be 44 ± 4.4.
3. When 120 ± 12 volts direct current (vdc) negative potential is applied to pin 3 and a positive potential is applied to pin 13, the "GO" indicator will light.
4. When 120 ± 12 vdc negative potential is applied to pin 2 and the positive potential is applied to pin 13, the "NO GO" indicator will light.
5. When a load consisting of a 22 ± 2 K ohms, 2 watts rating resistor and 1 microfarad (mfd) 300 vdc capacitor is connected between 2 and 6 and the handle is rotated a maximum of 3.5 revolutions per second (rps), the voltage across the load shall be a minimum of 120 vdc.
6. When direct current voltages listed in Table II are applied to pins 2 and 6 with positive potential on pin 2, the current shall vary between the limits specified.

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TABLE II

Direct Current Voltages and Current

VDC	Current
130 ± 5	0
175 ± 5	.2 to 2.2. milliamperes(ma)

Note: The same ammeter shall be used for both readings.

7. The resistance between pins 6 and 12 shall be less than 4 K ohms.

3.6 Dielectric strength. - The test set shall withstand a potential of 650 ± 50 volts root mean square (vrms) at 60 cycles per second (cps) (Utility Regulations acceptable; Local regulation ± 5 cps) between each ungrounded circuit, test set chassis, and unused connector pins without evidence of dielectric breakdown when tested as specified in 4.4.2. Upon completion of dielectric strength test, the test set shall comply 3.7.

3.7 Insulation resistance. - The insulation resistance between ungrounded circuits, test set chassis, and unused connector pins shall not be less than 30 meghoms when tested as specified in 4.4.3. and Table VI (see 4.4.2).

3.8 Operation. - After complying with 3.8.1 and 3.8.2, the test set shall be tested in accordance with 4.4.4 and both the "GO" and "NO GO" indicators shall light.

3.8.1 Test set. - When the handle of the test set is rotated at a maximum of 2.5 rps with pins 2 and 9 interconnected and pins 6 and 13 interconnected, the "GO" and "NO GO" indicators will light when tested as specified in 4.4.4.1.

3.8.2 Self test connector.

3.8.2.1 Circuit resistance. - The self test connector shall have a circuit resistance of no more than 0.1 ohms between pins 2 and 9 and between pins 6 and 13 when tested as specified in 4.4.4.2.

3.8.2.2 Insulation resistance. The insulation resistance shall not be less than 50 meghoms when a potential of 500 ± 50 vdc with current limited to 5 milliamperes is applied as follows: positive potential to pins 2 and 6 consecutively, negative potential to shell and all other pins except pins 9 and 13 consecutively when tested as specified in 4.4.3.

3.9 Vibration. - The test set enclosed in the transit case shall withstand vibrations in each of its three mutually perpendicular axes when tested as specified in 4.4.5. Upon completion of testing, the test set shall comply with 3.5, 3.7 and 3.8.

3.10 Temperature and humidity. The test set shall withstand exposure to the temperature and humidity cycle as specified in MIL-STD-170 when tested as specified in 4.4.6. After completion of the test, the test set shall comply with 3.5, 3.7, and 3.8.

3.11 Drop. - The test set enclosed in the transit case shall withstand the free fall drop tests as specified in MIL-P-116, Free Fall Drop Test, section 4. when tested as specified in 4.4.7. Upon completion of testing, the test set shall comply with 3.5, 3.7 and 3.8.

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3.12 Low temperature. - The test set shall withstand exposure to -65 ± 5 degrees fahrenheit. While at this temperature the test set shall comply with 3.5, 3.7 and 3.8. After completion of the test, when the test set has returned to a temperature of +80 ± 10 degrees fahrenheit, the test set shall comply with 3.5, 3.7 and 3.8. Testing shall be conducted as specified in 4.4.8.

3.13 Paint adhesion. - Paint adhesion shall comply with TT-C-490 requirement when tested as specified in 4.4.9.

3.14 Workmanship.

3.14.1 Plastic and metal characteristics. - All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.14.2 Threads. - Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.14.3 Burrs. - Parts shall be free from burrs.

3.14.4 Foreign matter. - Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.14.5 Cleaning. - The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

3.14.6 Electrical. - Cable assemblies and wires shall be dressed in such a manner as to prevent pinning or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection

4.2.1 Quantity to be submitted. - When the contract requires first article inspection, the contractor shall submit 2 complete assemblies (see dwg. 8844885), 2 subassemblies (see dwgs. 8834346, 8834408, 8834340, 8834489, 8834564, and 8834345 and 2 sets of parts, for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. - The first article inspection sample shall be subjected to the inspections and tests specified in Table III. Tests need not be conducted in order specified in Table III.

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TABLE III

First Article Inspections and Tests

Examination or Test	Contractor's Inspection Before Submission	Destructive	Non-Destructive	Parts	Sub-Assemb.	Assemb.
1. Classification of characteristics (see 4.3.2.1)	X		X	X	X	X
2. Circuits (see 3.5)	X		X		X	
3. Dielectric Strength(see 3.6)		X			X	
4. Insulation resistance (see 3.7)	X		X		X	
5. Operation (see 3.8)	X		X		X	
6. Vibration (see 3.9)	X		X		X	
7. Temperature and humidity (see 3.10)		X	X		X	
8. Drop (see 3.11)		X	X		X	
9. Low temperature (see 3.12)	X		X		X	
10. Paint adhesion (see 3.13)	X		X	X	X	X

4.2.3 Inspection equipment. - The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment speicified in 4.3.4.1.

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4.2.4 Rejection. - If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Product acceptance inspection.

4.3.1 Lot formation. - A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.1 Inspection lot. - An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection Government acceptance.

4.3.2 Product inspection examination. - Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table and inspected for all characteristics in the applicable classification. Any characteristic in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Table IV

Major Characteristic

Lot Size	Sample Size
	100% Inspection
1-155	155
156-200	175
201-250	190
251-300	200
301-350	210
351-400	220
401-450	225
451-500	230
501-550	235
551-600	240
601-650	243
651-700	247
701-750	250
751-800	254
801-900	258
901-1000	261
1001-1100	264
1101-1200	266
1201-1300	268
1301-1400	270
1401-1500	272
1501-1600	

Table V

Minor Characteristic

Lot Size	Sample Size
	100% Inspection
1- 25	25
26- 30	26
31- 35	27
36- 40	28
41- 45	29
46- 50	30
51- 55	31
56- 60	32
61- 65	33
66- 70	34
71- 75	35
76- 85	36
86- 95	37
96- 110	38
111- 130	39
131- 156	40
157- 190	41
191- 240	42
241- 340	43
311- 520	44
521- 1120	45
1121 and over	

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4.3.2.1 Classification of characteristics. - Critical, major, and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Coupling (see NAS 120106A);screw (see AN515-3R4); washer (see AN935-3); nut (see AN340-3); coupling and chain assembly (see NAS1201-C6B10A); connector electrical, screw (see MS35233-28 and MS35233-26) screw, drive (see AN535-0-3); insert heli-coil (see MS122116); resistor RC07GF223J (see MIL-R-118); resistor (see MS35043-119);packing "O" ring (see M528775); set screw (see MS51041-19 and MS51025-18); screw (see AN507C632R6) silicon zener regulator, 1N1779 (see dwg 8824530); silicon diode 1N647 (see dwg 8824531); lamp, neon glow (see dwg 8824548); terminal (see dwg 8824549); connector, receptacle, electrical (see dwg 8824556); connector, electrical (see dwg. 8834253); epoxy resin, top coat (see dwg 8834268); epoxy resin, putty blend (see dwg 8834269); plunger ball (see dwg 8834394) electrofilm No. 4856 (see dwg 8834440); barrier light (see dwg 8834470); electrofilm No99A (see dwg 8834523) covering details of dwgs 8834345, 8834346, 8834349, 8834408, and 8834564 as applicable.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Objective evidence (see 6.2) available for part in accordance with applicable standard or drawing	Visual
Minor:		
201.	Workmanship (see 3.14)	Visual-Manual

4.3.2.1.2 Connector assembly (see dwg 8834564 covering a detail of dwg 8834346).

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major:		
101.	Objective evidence (see 6.2) available for potting in accordance with dwg 8834564	Visual
102.	Wires not covered with potting	Visual
103.	Potting in areas which may impair electrical or mechanical functioning	Visual

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201. Workmanship (see 3.14.).....Visual

4.3.2.1.3 Yoke, crank (see dwg 8834386 covering a detail of dwg 8834345)

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Diameter of generator shaft hole	Gage
102.	Distance to flat on generator shaft hole, min	Gage
103.	Material (see 3.1)	Visual
Minor:		
201.	Workmanship (see 3.14)	Visual-Manual

4.3.2.1.4 Crank Assembly (see dwg 8834345 covering a detail of dwg 8834346).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Crank operates freely with respect to yoke through 180 degrees	Manual
102.	Handle stud tight	Manual
103.	Shaft detent hole in line with ball plunger	Manual
104.	Handle operates freely	Manual
105.	No part missing or improperly assembled	Visual
Minor:		
201.	Objective evidence (see 6.2) available for protective finish in accordance with dwg 8834345	Visual
202.	Workmanship (see 3.14)	Visual-Manual

4.3.2.1.5 Face plate, top (see dwg 8834471) and face plate, bottom (see dwg 8834472) covering details of dwg 8834480.

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Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Material (see 3.1)	Visual
Minor: None defined.		

4.3.2.1.6 Module assembly indicator light (see dwg 8834340 covering a detail of dwg 8834408) before potting.

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Lamp incorrectly assembled	Visual
102.	Lacquer missing	Visual
103.	Lacquer on surfaces x and z	Visual
104.	Terminals loose	Manual
105.	Materials (see 3.1)	Visual
Minor:		
201.	Workmanship (see 3.14.)	Visual-Manual

4.3.2.1.7 Module assy, indicator light (see dwg 8834340 covering a detail of dwg 8834408)

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Potting material (see 3.1)	Visual
102.	Proper potting	Visual
Minor: None defined.		

4.3.2.1.8 Module assembly, component, electrical (see dwg 8834408 covering a detail of dwg 8834346) before potting.

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Loose wires length specified	Gage

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Minor:
201. Marking on wire leads.....Visual
202. Workmanship (see 3.14.).....Visual-Manual

4.3.2.1.9 Module assembly, component, electrical (see dwg 8834408 covering a detail of dwg 8834346).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Potting does not extend beyond rear diameter of module indicating light	Visual
102.	Proper potting	Visual
103.	Potting material (see 3.1)	Visual
Minor:		
201.	Workmanship (see 3.14.)	Visual-Manual

4.3.2.1.10 Housing, test set, atomic weapon T4040 (see dwg 8834349 covering a detail of dwg 8834346

Categories	Chacteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Symmetry of holes	Gage
102.	Generator clearance hole diameter, min	Gage
103.	Length and width of generator mounting cavity, min	Gage
104.	Depth to mounting surface for module assembly	Gage
105.	Thickness of casting at connector plate mounting surface, min	Gage
106.	Contour of housing	Visual
107.	Small diameter of module assy mounting hole	Gage
108.	Large diameter of module assy mounting hole	Gage
109.	Concentricity of large diameter of module assy hole	Gage
110.	Material (see 3.1)	Visual
Minor:		
201.	Objective evidence (see 6.2) available for protective finish in accordance with dwg 8834349	Visual
202.	Workmanship (see 3.14)	Visual-Manual

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4.3.2.1.11 Housing, test set, atomic weapon T4090 (see dwg 8834349 covering a detail of dwg 8834346)

Categories	Characteristics	Method of Inspection
Critical: None defined		
Major: None defined.		
Minor:		
201.	No paint missing	Visual
202.	No paint in areas designated on dwg	Visual

4.3.2.1.12. Plate, cover, generator (see dwg 8834337 covering a detail of dwg 8834346) before painting.

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Location of 2 small holes	Gage
102.	Diameter of 2 small holes	Gage
103.	Diameter of countersink	Gage
104.	Diameter of large hole	Gage
105.	Location of large hole	Gage
106.	Thickness, min	Gage
107.	Material (see 3.1)	Visual
Minor:		
201.	Objective evidence (see 6.2) available for protective finish in accordance with dwg 8834337	Visual
202.	Workmanship (see 3.14)	Visual-Manual

4.3.2.1.13 Plate, Cover, generator (see dwg 8834337 covering a detail of dwg 8834346).

Categories	Characteristics	Method of Inspection
Critical; None defined.		
Major: None defined.		
Minor:		
201.	No paint missing including bare spot	Visual

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202. Workmanship (see 3.14.).....Visual

4.3.2.1.14. Plate connector (see dwg 8834347 covering a detail of dwg 8834346).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	No paint on internal surfaces or surfaces marked "X"	Visual
Minor:		
201.	No paint missing	

4.3.2.1.15 Identification plate test set atomic weapon T4090 (see dwg 8834465 covering a detail of dwg 8834346)

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Marking	Visual
Minor:		
201.	Workmanship (see 3.14)	Visual

4.2.2.1.16. Test set, Atomic Weapon T-4090 (see dwg 8834346) before assembly of module assembly and plate, connector.

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Generator (see 3.2.1)	Visual
102.	Torque on generator assembly screws	Gage
103.	Cable correctly laced	Visual
104.	Wire not too short to permit assembly of spare parts	Visual
105.	Epoxy resin putty on generator terminals	Visual
106.	No part missing or incorrectly assembled	Visual
107.	Crank turns freely	Manual
108.	Crank functions properly	Manual
109.	Materials (see 3.1)	Visual

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Minor:
201. Workmanship (see 3.14.).....Visual-Manual

4.3.2.1.17 Test set, Atomic Weapon T-4090 (see dwg 8834346) before filling connector well with epoxy resin.

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:
101. Wiring in accordance with dwgs 8834409 and 8834410.....Visual
102. Torque on connector plate, screws.....Gage
103. Any part missing or incorrectly assembled.....Visual-Manual
104. Soldering in accordance with MIL-S-60902(MU).....Visual-Manual
106. Materials (see 3.1).....Visual

Minor:
201. Workmanship (see 3.14.).....Visual-Manual

4.3.2.1.18 Test Set, Atomic Weapon, T-4090 (see dwg 8834346).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:
101. Screws in cover plate and crank assembly tight.....Manual
102. No pad loose.....Visual-Manual
103. No part missing or incorrectly assembled.....Visual

Minor:
201. Lacquer present on identification plate.....Visual
202. Marking .....Visual
203. No paint missing, including bare spot.....Visual
204. Serial number in accordance with 3.4.....Visual
205. Workmanship (see 3.14).....Visual-Manual

4.3.2.1.19 Test set atomic weapon: T-4090 w/case (see dwg 8834979 and 5.1) prior to closing case.

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Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Case (see 3.2.2)	Visual
102.	Any component missing	Visual
Minor:		
201.	Workmanship (see 3.14)	Visual/Manual

4.3.2.1.20 Test set atomic weapon: T-4090 w/case (see dwg 8834979 and 5) after closing case.

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Car seal present and properly secured	Visual-Manual
Minor:		
201.	Marking	Visual
202.	No paint missing including bare spot	Visual
203.	Serial number in accordance with 3.4	Visual
204.	Workmanship (see 3.14.)	Visual

4.3.2.1.21 Box, pkg. ammo, for test set: T4090 (see dwg 8844885 and 5) before closing cover.

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Transit case tight in box	Manual
102.	Proper use of fillers	Visual
103.	Box in accordance with MIL-B-2427	Visual
Minor:		
None defined.		

4.3.2.1.22 Box, pkg, ammo, for test set T-4090 (see dwg 3344885 and 5 after closing cover.

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Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	No strapping missing, broken, or loose	Visual-Manual
102.	No hardware missing, broken, or loose	Visual-Manual
103.	No board broken or split	Visual
104.	Car seal present, sealed	Visual
105.	D. O. D. symbol present and identifiable	Visual
Minor:		
201.	Hardware properly fastened	Visual/Manual
202.	Steel strapping, properly located	Visual
203.	Marking	Visual
204.	Serial number in accordance with 3.4	Visual
205.	Workmanship (see 3.14.)	Visual-Manual

4.3.3 Testing. - The contractor shall perform all the tests specified in this paragraph. Tests do not have to be performed in the order specified.

Test		Test Classification
a.	Circuits	Major
b.	Insulation resistance	Major
c.	Operation	Major
d.	Vibration	Major
e.	Module assembly indicator light	Major
f.	Aging neon lamps	Major
g.	Firing voltage neon lamps	Major
h.	Paint adhesion	Minor

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Circuits (see 3.5), insulation resistance (see 3.7.) and operation (see 3.8.).- Each test set in the lot shall be subjected to these tests. See applicable paragraph in 4.4. If any test set fails to comply with the applicable requirement, it shall be classified defective and removed from the lot.

4.3.3.2 Vibration (see 3.9.) - One test set for every fifty (50) or less manufactured shall be tested in accordance with 4.4.5. If any test set fails to comply with 3.9., the lot shall be rejected.

4.3.3.3 Module assembly indicator light (see dwg 8834340). - The indicator lights on each module assembly shall be tested a s specified in 4.4.10. If any indicator light is shorted, the assembly shall be rejected and removed from the lot.

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4.3.3.4 Aging Neon lamps and firing voltage (see dwg 8824548). - Each neon lamp shall be aged and checked for firing voltage as specified on the drawing. If any lamp fails during the aging process or fails to comply with firing voltage drawing requirements, it shall be classified defective and removed from the lot.

4.3.3.5 Paint adhesion (see 3.13 and dwgs 8834386, 8834388, 8834341, 8834349, 8834337, 8834347, and 8834346). Each part shall be tested as specified in 4.4.9. If any part fails to comply with 3.13, it shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). EL's for assemblies are listed on the Equipment Tabulation, ET-8834346. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment. The contractor shall obtain approval of such designs, operating instructions, and special calibration instructions as specified in 6.3 prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired.
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL.

4.3.4.2 Minor characteristics and tests. - Acceptance inspection equipment for minor characteristics is not specified on EL's. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. - The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Circuits (see 3.5). - Check the circuits with test equipment listed on the EL in accordance with Table I (see 3.5).

4.4.2 Dielectric strength (see 3.6). - With an approved dielectric strength tester connect a potential of 650 ± 50 v,r,m,s, at 60 ± 5 c,p,s, for a minimum period of 15 seconds between connector pins as specified in Table VI.

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TABLE VI

Dielectric Strength and Insulation Resistance

Positive Potential to Pin	Negative Potential to All Other Pins and Connector Shell Except Pins
6	2, 3, 9
11	2, 3, 9
12	2, 3, 9
13	2, 3, 9

After completion of the test, comply with 3.7.

4.4.3 Insulation resistance (see 3.7 and 3.8.2.2). - Measure insulation resistance with an approved insulation resistance tester capable of delivering 500 ± 50 volts direct current, limited to 5 miliamperes.

4.4.4 Operation (see 3.8). - After completion of 4.4.1 and 4.4.2 tests, the self test connector shall be assembled to the test set, and the handle rotated at a maximum of 2.5 vps.

4.4.4.1 Test Set (see 3.8.1). - Interconnect pins 2 and 9 and pins 6 and 13, respectively, and rotate the hand crank at a maximum of 2.5 rps.

4.4.4.2 Self test connector circuit resistance (see 3.8.2.1). - The circuit resistance shall be measured with approved low resistance measuring equipment.

4.4.5 Vibration (see 3.9).

4.4.5.1 Shaker-Fixture-Component Resonance Test. - Prior to performance of the component vibration test, subject each component to vibration in each of its three major mututally perpendicular axes throughout the frequency range from 5-55 cps unless this has already been done with an identical component and an identical fixture on the same machine to be used for the component vibration test. Attach component to test fixture by means of the attachment points used to mount the component in service in such a manner as to prevent additional stiffness or restraint to the component other than at the attachment point. Monitor the vibration response continuously during this test at one of the component-to-fixture attachment points. Note resonant frequencies, band width and amplification of all resonances. Amplitude readings perpendicular to the direction of excitation shall be no greater than the maximum level in the direction of excitation. Alter the component-fixture-shaker combination until these conditions are satisified. Control frequency sweep manually to insure that all resonances observed are fully excited. Vary amplitude with the frequency such that the maximum amplitude at the attachment points does not exceed the following:

Frequency	Amplitude
5-55	.03 - Peak to Peak

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If any alterations are made to the machine fixture or component, or, if a different machine or fixture is used after the above test has been made, repeat this test, incorporating the new conditions, prior to component testing.

4.4.5.2 Component test (see 3.9). - With the test set, enclosed in its carrying case in its intended carrying condition and mounted to the vibration machine in the manner determined by the fixture resonance test (4.4.5.1), vibrate the test set in each of 3 mutually perpendicular axes at the frequency, time and amplitude ranges specified below. After completing the vibration runs, inspect the test set for compliance with 3.5, 3.7, and 3.8.

Frequency (CPS)	Sweep Time (Minutes)	Double Amplitude (Inches)	Number of Sweeps
5-55-5	1 ± .2	.06 ± .006	15

4.4.6 Temperature and humidity (see 3.10). - Place the test sets in a temperature humidity test chamber and subject them to the test cycle specified in MIL-STD-170. Make certain that a connector cap is in place on each connector. Maintain the specified temperature within plus or minus 5 degrees fahrenheit. Control the relative humidity with distilled or demineralized water having a PH value between 6.5 and 7.5 at 77 degrees fahrenheit. Adjust PH values with chemically pure hydrochloric acid or sodium hydroxide. Upon completion of the test cycle, allow the test set to return to ambient temperature without use of forced cooling. Remove connector caps and remove excessive moisture from the test set with dry, filtered compressed air. Inspect the test set for compliance with 3.5, 3.7, and 3.8.

4.4.7 Drop test. - Subject the test set, enclosed in its carrying case in the intended carrying condition, to the free fall drop test specified in MIL-P-116, then inspect the test set for compliance with 3.5, 3.7, and 3.8.

4.4.8 Low temperature. - The test set with a connector cap in place on the connector shall be placed in a test chamber and the temperature lowered to -65 ± 5 degrees fahrenheit at ambient humidity for a period of 4 hours or until the temperature stabilizes (see 6.4). While at this temperature the test set shall comply with 3.5, 3.7, and 3.8. Upon completion of the test cycle allow the test set to return to ambient temperature (80 ± 0 degrees fahrenheit), without the use of forced heating. Remove connector caps and remove moisture from the test set connector with dry, filtered, compressed air. Inspect the test set for compliance with 3.5, 3.7, and 3.8.

4.4.9 Paint adhesion. - Paint adhesion shall be tested as specified in TT-C-490.

4.4.10 Module assembly indicator light (see dwg 8834340 and 4.3.3.3). - Each indicator light shall be checked with a vacuum tube voltmeter (VTM) set to measure resistance. If the resistance measurement is less than 10 megohms, the lamp will be considered shorted and the assembly rejected.

5.0 PREPARATION FOR DELIVERY

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5.1 Preservation and packaging. - Preservation and packaging shall be level A or C as specified (see 6.1 of SQAP - Appendix I).

5.1.1 Level A. - The test set shall be packaged in accordance with dwgs 8834979 and 8844885 using the specified moisture barrier.

5.1.2 Level C. - The test set shall be packaged in accordance with dwg 8844885 except that the moisture barrier may be omitted.

5.2 Packing

5.2.1 Level A and C. - The test set shall be packed in accordance with dwgs 8834979 and 8844885.

5.3 Marking. - Marking shall be as specified in accordance with dwgs 8834979 and 8844885, as applicable.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.2 Objective evidence. - Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment. - The contractor shall obtain approval of equipment designs, operating instructions, and calibration instructions from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-NR4.

6.4 Temperature stabilization. - The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. - This specification includes the requirements of Purchase Description PA-PD-1869 dated 14 December 1965.

Custodian:
Army (MU)

Preparing Activity:
Army (MU)

Project No. 1190-A002

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DEPARTMENT OF THE ARMY

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OFFICIAL BUSINESS

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-NR
Dover, New Jersey 07801

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-T-50278(MU) Test Set Atomic Weapon: T4090

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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